

Lab: Bio-Inspired Navigation and Foraging

Objective

Design navigation and foraging algorithms inspired by biological strategies. You will analyze how animals plan routes, forage optimally, and make decisions under uncertainty, then implement bio-inspired planning algorithms for autonomous robots.

Prerequisites

- Completion of the corresponding module.md lecture material
- Familiarity with Active Inference concepts (generative models, free energy, prediction errors)
- Basic pseudocode and diagram skills

Part 1: Path Integration and Dead Reckoning

Analyze biological path integration (desert ant navigation) as Active Inference planning:

1. How does the desert ant maintain a home vector using optic flow and step counting?
2. Model this as a generative model that predicts position from self-motion cues.
3. How does the ant decide when to switch from exploration to homing?
4. Design a robotic path integration system inspired by ant navigation for GPS-denied environments.

Write your response here

Part 2: Optimal Foraging Theory

Apply optimal foraging theory to robotic task allocation:

1. The marginal value theorem: when should a robot leave a depleting resource patch?
Frame as expected free energy evaluation.
2. Design a foraging policy where robots exploit known reward locations while exploring for new ones.
3. How does the explore-exploit trade-off in foraging map to pragmatic vs. epistemic value in Active Inference?

Write your response here

Part 3: Cognitive Maps and Spatial Planning

Compare biological cognitive maps with robotic planning:

1. How do rats use hippocampal replay to plan novel shortcuts they have never traveled?
2. Design a robotic planning system that uses experience replay to discover new routes.
3. How does the concept of a cognitive map differ from a metric map, and what are the planning implications?

Write your response here

Part 4: Bio-Inspired Planning Comparison

Compare planning strategies:

Feature	Desert Ant	Honeybee	Rat
Spatial representation			
Planning horizon			
Optimality			
Novel route discovery			

Write your response here

Summary Table

Concept	Classical Robotics	Active Inference	Your Design
Core mechanism			
Uncertainty handling			
Adaptation			
Key advantage			

References

- Lanillos, P., et al. (2021). Active Inference in Robotics and Artificial Agents. *Frontiers in Neurorobotics*.
- Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference*. MIT Press.
- Pio-Lopez, L., Nizard, A., Friston, K., & Pezzulo, G. (2016). Active Inference and robot control. *Journal of the Royal Society Interface*, 13(122).